Pollutants

SWAT+ Documentation

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Purpose

The pesticide module simulates user-listed pesticides across foliage, soil, channel water, benthic sediment, and aquifers. The set of pesticides simulated is declared in constituents.cs, which populates `cs_db%num_pests`. The files below load the parameter database, the per-object initial conditions, and the parent-daughter metabolite chain.

Activation

The pesticide parameter database `pesticide.pes` is referenced from `file.cio` as `in_parmdb%pest` (default `pesticide.pes`). The initial-condition files are referenced from `file.cio` via `in_init%pest_soil` (default `pest_hru.ini`) and `in_init%pest_water` (default `pest_water.ini`).

`pest_parm_read` is called from `proc_db.f90`. `pest_metabolite_read` and `pest_hru_aqu_read` are called from `proc_read.f90`. `pest_cha_res_read` is called later from `main.f90`.

Files

pesticide.pes

Pesticide parameter database. One row per pesticide.

- Reader: src/pest_parm_read.f90
- Type: `pestdb(0:pestparm)` (`type pesticide_db`) in src/pesticide_data_module.f90. The reader also fills `pestcp(ip)%decay_f`, `decay_s`, `decay_a`, `decay_b` from the half-life columns as `exp(-0.693 / hlife)`.
- Format: title line, header line, then one line per pesticide.

#	Field	Units	Description
1	name	-	pesticide name
2	koc	mL/g	soil adsorption coefficient normalized for soil organic carbon

#	Field	Units	Description
3	washoff	none	fraction of pesticide on foliage washed off by a rainfall event
4	foliar_hlife	days	half-life on foliage
5	soil_hlife	days	half-life in soil
6	solub	mg/L	solubility in water
7	aq_hlife	days	aquatic half-life
8	aq_volat	m/day	aquatic volatilization coefficient
9	mol_wt	g/mol	molecular weight, used for mixing velocity
10	aq_resus	m/day	aquatic resuspension velocity for sorbed pesticide
11	aq_settle	m/day	aquatic settling velocity for sorbed pesticide
12	ben_act_dep	m	depth of the active benthic layer
13	ben_bury	m/day	burial velocity in benthic sediment
14	ben_hlife	days	half-life in benthic sediment
15	pl_uptake	none	fraction taken up by plant
16	descrip	-	description string

pest_hru.ini

Initial pesticide concentrations for HRU soils and plants. One block per record.

- Reader: src/pest_hru_aqu_read.f90
- Type: `pest_soil_ini` (`type cs_soil_init_concentrations`) in src/constituent_mass_module.f90. Fields: `name`, `soil(num_pests)` (ppm), `plt(num_pests)` (ppm).
- Format: one title line, then per record: a header line, a name line, then `num_pests` lines each containing `pest_name, soil_value, plt_value` in the order declared in `constituents.cs`.

The same reader populates `cs_pest_solsor(num_pests)` (allocated only).

pest_water.ini

Initial pesticide concentrations for channels and reservoirs (water column and benthic).

- Reader: src/pest_cha_res_read.f90
- Type: `pest_water_ini` (`type cs_water_init_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `water(num_pests)`, `benthic(num_pests)`. Record names are stored in `pest_init_name`.
- Format: title line, header line, then per record: name line, then `num_pests` pairs of lines (water value, benthic value). The values are then read again on the second pass as a single record line of `name, water(:), benthic(:)`.

pest_metabolite.pes

Parent-daughter metabolite chain. Filename is hard-coded in the reader (not driven by `file.cio`).

- Reader: src/pest_metabolite_read.f90
- Target: `pestcp(ip)%daughter(num_metab)` (`type daughter_decay_fractions`) and `pestcp(ip)%num_metab` in `src/pesticide_data_module.f90`. The daughter `%num` field is set by cross-walk against `cs_db%pests` so the metabolite must also appear in `constituents.cs`.
- Format: title line, header line, then per parent: one line `parent_name num_metab`, followed by `num_metab` lines of `daughter_name, foliar_fr, soil_fr, aq_fr, ben_fr`. The parent name must match an entry in `pesticide.pes`.

Field	Units	Description
name	-	daughter pesticide name
foliar_fr	0-1	fraction of parent on foliage that degrades to this daughter
soil_fr	0-1	fraction of parent in soil that degrades to this daughter
aq_fr	0-1	fraction of parent in aquatic phase that degrades to this daughter

Field	Units	Description
ben_fr	0-1	fraction of parent in benthic phase that degrades to this daughter

Related

- constituents.cs. Declares the pesticide list and sets cs_db%num_pests.
- pathogen module. Same per-object initial-condition pattern.
- metals module. Same per-object initial-condition pattern.

Purpose

The pathogen module simulates user-listed pathogens across soil, foliage, channel water, benthic sediment, and aquifers. The set of pathogens simulated is declared in constituents.cs, which populates cs_db%num_paths. The files below load the parameter database and the per-object initial conditions.

Activation

The pathogen database pathogens.pth is referenced from file.cio as in_parmdb%pathcom_db (default pathogens.pth). The initial-condition files are referenced via in_init%path_soil (default path_hru.ini) and in_init%path_water (default path_water.ini).

path_parm_read is called from proc_db.f90. path_hru_aqu_read is called from proc_read.f90. path_cha_res_read is called later from main.f90.

Files

pathogens.pth

Pathogen parameter database. One row per pathogen.

- Reader: src/path_parm_read.f90
- Type: path_db(0:path) (type pathogen_db) in src/pathogen_data_module.f90.
- Format: title line, header line, then one line per pathogen.

#	Field	Units	Description
1	pathnm	-	pathogen name
2	do_soln	1/day	die-off factor in soil solution
3	gr_soln	1/day	growth factor in soil solution
4	do_sorb	1/day	die-off factor sorbed to soil particles
5	gr_sorb	1/day	growth factor sorbed to soil particles
6	kd	none	partition coefficient between solution and sorbed phase in surface runoff
7	t_adj	none	temperature adjustment factor for die-off and growth
8	washoff	none	fraction on foliage washed off by a rainfall event
9	do_plnt	1/day	die-off factor on foliage
10	gr_plnt	1/day	growth factor on foliage
11	fr_manure	none	fraction of manure with active colony forming units
12	perco	none	percolation coefficient, ratio of solution bacteria in the surface layer
13	det_thrshd	cfu/m^2	detection threshold below which soil pathogen levels are zeroed
14	do_stream	1/day	die-off factor in streams
15	gr_stream	1/day	growth factor in streams
16	do_res	1/day	die-off factor in reservoirs
17	gr_res	1/day	growth factor in reservoirs

#	Field	Units	Description
18	conc_min	-	minimum pathogen concentration

path_hru.ini

Initial pathogen concentrations for HRU soils and plants. One block per record.

- Reader: src/path_hru_aqu_read.f90
- Type: `path_soil_ini` (`type cs_soil_init_concentrations`) in src/constituent_mass_module.f90. Fields: `name`, `soil(num_paths)`, `plt(num_paths)` in cfu/m^2.
- Format: one title line, then per record: a header line, a name line, then one line containing `name, soil(:), plt(:)` for all pathogens in `constituents.cs` order.

The same reader allocates `cs_path_solsor(num_paths)`.

path_water.ini

Initial pathogen concentrations for channels and reservoirs (water column and benthic).

- Reader: src/path_cha_res_read.f90
- Type: `path_water_ini` (`type cs_water_init_concentrations`) in `src/constituent_mass_module.f90`. Fields: `name`, `water(num_paths)`, `benthic(num_paths)`. Record names are stored in `path_init_name`.
- Format: title line, header line, then per record: name line, then `num_paths` pairs of lines (water value, benthic value). On the second pass the reader reads a single line `name, water(:), benthic(:)` per record.

Related

- constituents.cs. Declares the pathogen list and sets `cs_db%num_paths`.
- pesticide module. Same per-object initial-condition pattern.
- metals module. Same per-object initial-condition pattern.

Purpose

The heavy metals module simulates user-listed metals across HRU soils and plants. The set of metals simulated is declared in constituents.cs, which populates `cs_db%num_metals`. Only initial soil and plant concentrations are loaded by a dedicated reader; no separate metal parameter database is read by `hmet_hru_aqu_read`.

Activation

`in_parmdb%hmetcom_db` is defined in src/input_file_module.f90 with default `metals.mtl`. No reader in `src/` opens this filename, so the parameter database is declared in `file.cio` but not loaded.

The initial-conditions file is referenced from `file.cio` via `in_init%hmet_soil` (default `hmet_hru.ini`). `hmet_hru_aqu_read` is called from `proc_read.f90`.

Files

hmet_hru.ini

Initial metal concentrations for HRU soils and plants.

- Reader: src/hmet_hru_aqu_read.f90
- Type: `hmet_soil_ini` (`type cs_soil_init_concentrations`) in src/constituent_mass_module.f90. Fields: `name`, `soil(num_metals)` (ppm), `plt(num_metals)` (ppm).
- Format: one title line, then per record: a header line, a name line, then `num_metals` pairs of lines (`metal_name, soil_value` then `metal_name, plt_value`) in `constituents.cs` order.

The same reader allocates `cs_hmet_solsor(num_metals)`.

metals.mtl

Declared as `in_parmdb%hmetcom_db` in `input_file_module.f90` (default name `metals.mtl`). No reader in the current source opens this file. The file is reserved but inert.

Related

- `constituents.cs`. Declares the metal list and sets `cs_db%num_metals`. The metals block in `constit_db_read` is read but its name-database cross-walk is commented out, so listed metals only contribute the count.
- `pesticide module`. Same per-object initial-condition pattern.
- `pathogen module`. Same per-object initial-condition pattern.